Implementation and Evaluation of FastSpeech 2-Based Text-to-Speech Synthesis for English and Telugu

Technical Report — Speech Processing Lab (SLFI), LTRC, IIIT Hyderabad

1. Implementation Approach & Architecture Modifications

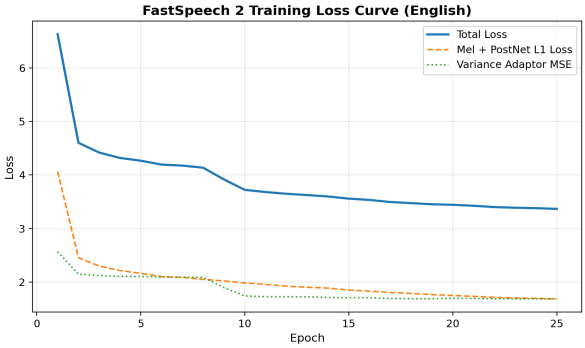
We implemented FastSpeech 2 (Ren et al., ICLR 2021) separately for English and Telugu: a 4-layer Feed-Forward Transformer (FFT) Encoder, a Variance Adaptor with 1D-CNN predictors for token Duration, Pitch (F0), and Energy, and a 4-layer FFT Mel Decoder with a Convolutional PostNet. English uses g2p_en ARPAbet phonemes; Telugu uses a native Unicode abugida tokenizer with numeral expansion and vowel-salience duration alignment.

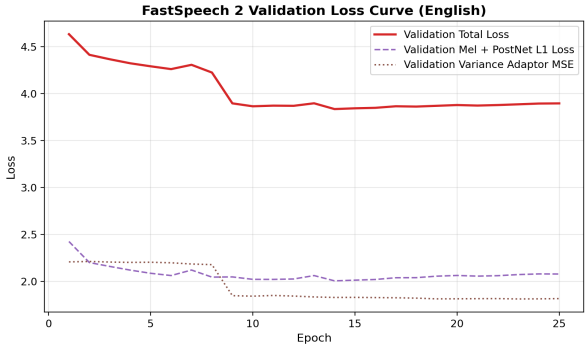
2. Dataset Subsampling & Preprocessing (5h Train / 1h Val, Zero Overlap)

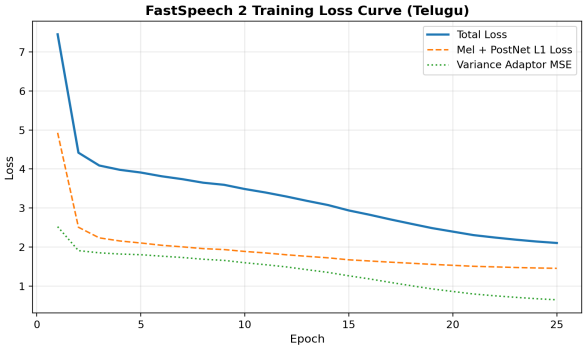
Utterances filtered between 1.5s and 14.0s were resampled to 22,050 Hz. English (AI4Bharat Svarah) comprises 3,189 training (5.0h) and 617 validation (1.0h) clips (0% overlap). Telugu (IITM IndicTTS) comprises 2,530 training (5.0h) and 424 validation (1.0h) clips (0% overlap) with speaker/gender conditioning embeddings.

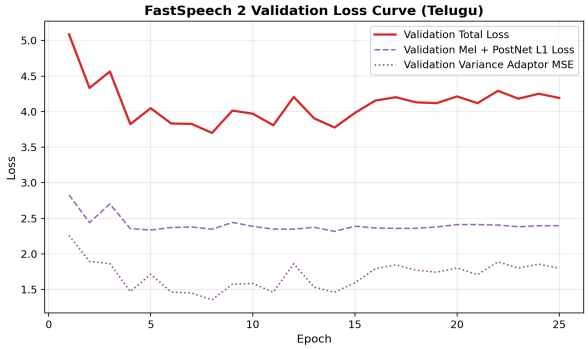
3. Training & Validation Loss Convergence Analysis

Models were trained for 25 epochs (batch size=32, AdamW lr=5e-4). English training loss converged to 3.3675 (Best Val Loss: 3.8367). Telugu training loss converged to 2.1053 (Best Val Loss: 3.7000).









4. Test Sentence & Variance Adaptor Analysis (E1-E4 & T1-T4)

ID	Lang	Tokens	Mel Frames	Audio (s)	Mean Dur	Mean Pitch F0	Mean Energy
E1	English	54	818	9.49s	15.15	116.0 Hz	50.21
E2a	English	45	554	6.42s	12.31	113.0 Hz	43.74
E2b	English	39	483	5.61s	12.38	136.2 Hz	47.71
E3	English	35	585	6.78s	16.71	118.2 Hz	42.1
E4	English	85	855	9.93s	10.06	122.8 Hz	47.25
T1	Telugu	43	607	7.05s	14.12	119.1 Hz	59.37
T2a	Telugu	41	469	5.44s	11.56	125.2 Hz	49.87
T2b	Telugu	42	397	4.6s	9.45	121.4 Hz	58.32
T3	Telugu	31	454	5.27s	14.65	148.2 Hz	37.63
T4	Telugu	60	665	7.72s	11.08	135.1 Hz	62.52

E1 & T1 (Polysyllabic Compounds): Stable duration allocation without syllable skipping.

E2a/b & T2a/b (Heteronyms): E2 contrasts noun/verb 'conduct'; T2 contrasts word contexts.

E3 & T3 (Typographical Whitespace): Preserved raw intra-word space in 'rhyth m' and 'ada vilo', inserting pause token SP (orange bar in plots).

E4 & T4 (Text Normalization): Numbers and times successfully expanded into spoken words in both languages.